Novel Bis- and Mono-Xylosides Synthesis and Biological Action

Aidan Yu

Department of Medicinal Chemistry, University of Utah College of Pharmacy

Author Note

Aidan Yu https://orcid.org/0009-0005-9589-5314

The authors have no conflicts of interest to declare. Research involving non-human vertebrates or human subjects was conducted under the supervision of an experienced teacher or researcher, and state and federal regulatory guidance was followed applicable to the humane and ethical conduct of such research.

Correspondence concerning this paper should be addressed to Aidan Yu, Department of Medicinal Chemistry, University of Utah, 30 South 2000 East Salt Lake City, UT 84112-5820. Email: U0901397@umail.utah.edu

Abstract

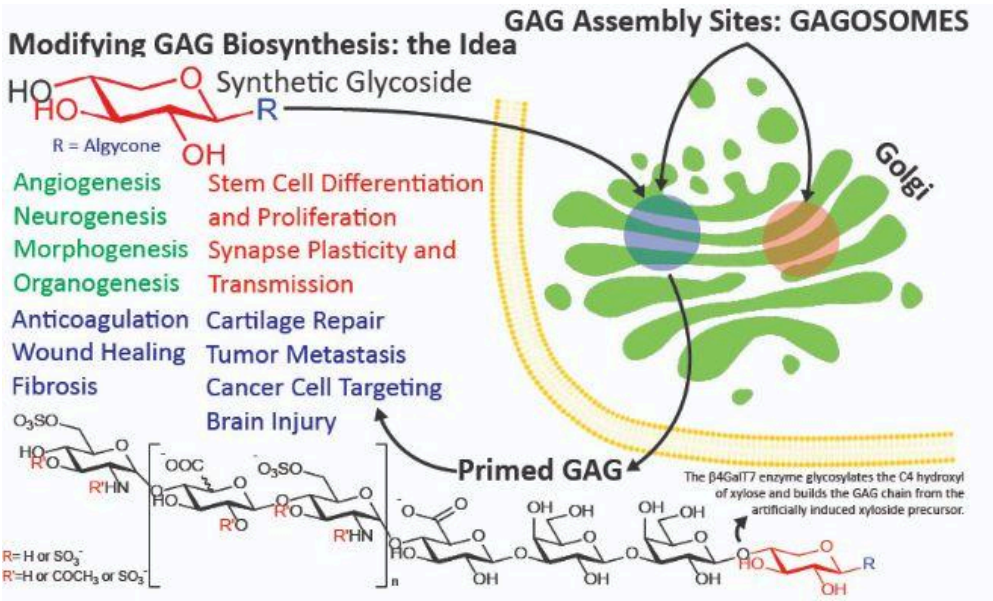
Complex sugars known as Glycosaminoglycans (GAGs) play a vital role in nearly all biological processes, from wound healing to the progression of diseases such as cancer and Alzheimer's. They are among the most complex, least understood, and most important biological macromolecules in all mammals. Despite the immense therapeutic potential of GAGs, their sheer complexity and related synthetic difficulties have greatly hindered their role in medicinal applications. Additionally, the absence of methods to study the spatial and temporal distribution of GAGs and recapturing their structures from in vivo significantly hampers our understanding of the mechanisms underlying glycan-based drug actions. I present my development of new methodologies for the efficient glycosylation of sugars. These methodologies were employed for the synthesis of novel xyloside bis- and mono-scaffolds. The bis-xyloside scaffolds were tested for fibroblast growth factor(FGF) signaling and most notably, the method allowed for these bis-xyloside scaffolds to have a fluorophore attached to them to elude their mechanisms of action. This work addresses many shortcomings related to glycomics and glycan-based drug design. These developments offer new methods of glycan synthetic chemistry, notably one that accelerates glycosylation by over 1000x, and also glycan-based approaches for developing new, more effective disease treatments specifically, a new therapeutic for mucopolysaccharidosis is being developed.

Keywords: Xylosides, Glycosaminoglycan(GAG), Glycan, Fibroblast Growth Factor (FGF) Signaling, Zebrafish Embryonic Development

Introduction

Glycosaminoglycans (GAGs) spearhead a wide variety of biological processes. Endogunosly, GAGs are found in the proteoglycan form. This is a large molecule where GAG chains come off a central protein scaffold like a bottle brush. The artificial regulation of GAG's priming mechanisms and structures is crucial, as it holds significant potential in treating various diseases. This can be done with synthetic xylosides. Xylose is a carbohydrate in proteoglycans found in mammalian cells and is the second most abundant carbohydrate after cellulose. Xylosides imitate xylosylated serine residues found in proteoglycan core proteins, essentially behaving like a false acceptor to the priming of GAG chains. This leads to the building of GAG chains off of the introduced xyloside. This steers the assembly process away from the natural core proteins and results in the inhibition of proteoglycan formation. Instead, free-floating, individual GAG chains are created. Cells exposed to xylosides release individual glycosaminoglycan chains and gather proteoglycans with shortened chains.

Figure 1: Modifying GAG Biosynthesis: the Idea



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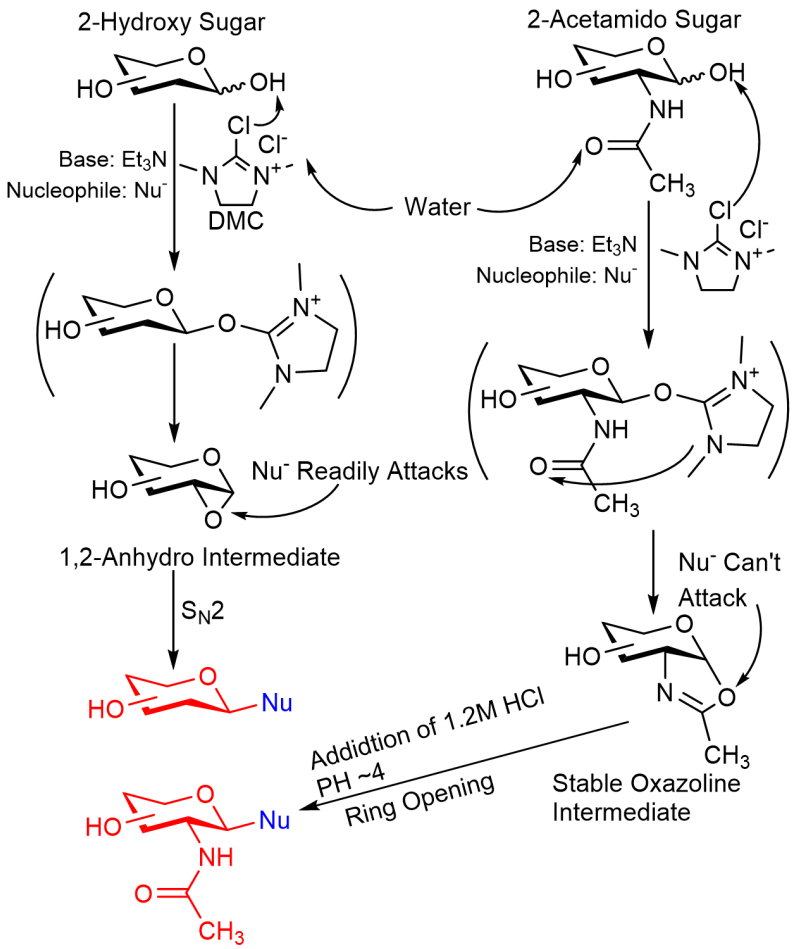
One-Pot Protection Group Free Glycosylation of Xylosides:

Current procedures for synthesizing xylosides require multiple reactions, the use of highly toxic chemicals, and are performed in organic solvents that are not environmentally friendly (Figure 2). This process is not only time-consuming, taking weeks to complete, but also poses potential biological and chemical hazards. A new, efficient process using 2-chloro-1,3-dimethylimidazolinium chloride (DMC) was developed here to address the challenges of traditional glycoside synthesis. This approach allows for the synthesis of glycosides taking only 30 minutes, in a single step, and without using protection groups. Notably, while traditional organic chemistry relies on organic solvents that greatly contribute to the global warming crisis, this method is carried out in water, eliminating the environmental toll of harmful chemicals. This process is also widely applicable to a wide range of oligosaccharides. 2-Chloro-1,3-dimethylimidazolinium chloride (DMC) activates the anomeric center of unprotected xylose. This property enables unprotected xylose to undergo anomeric substitution with nucleophiles, offering efficient reaction routes to xylosides, and

Figure 2: Glycosylation with DMC



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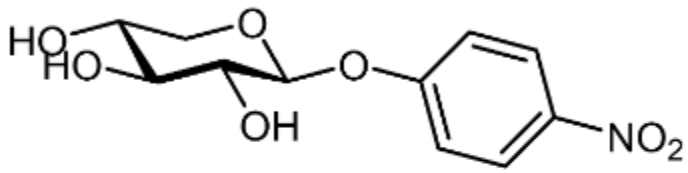
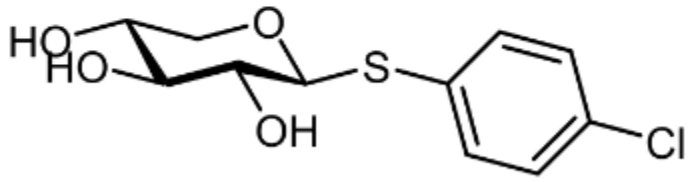
eliminating the need for traditional protecting group manipulations. Here DMC chemistry was used to efficiently synthesize xylosyl azides which were then converted into click xylosides in large quantities in the development of a therapeutic for Mucopolysaccharidosis. Now, DMC chemistry is directly compatible with click chemistry and is highly conducive to accelerating the development of glycan-based therapeutics of many types.

Table 1: Structures of Xylosides a and b

a	
b	

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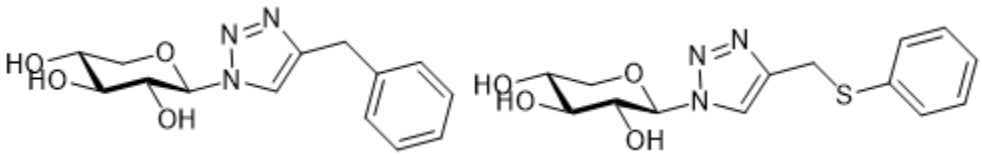
GAG priming of xylosides was tested in vitro with CHO pgsA-745 cells. Xyloside(a) primes GAGs, however, due to the oxygen linkage, it is metabolically unstable and can hydrolyze within the cell releasing toxic nitrophenyl. To overcome this challenge, xyloside(b) was synthesized, with a sulfur linkage. This linkage is more stable within biological environments and hydrolyzes much less readily making it better for GAG priming. Yet, it doesn't have high biological activity and doesn't exhibit strong decoy properties. Despite these issues, these molecules serve as a basis for a potential treatment for Mucopolysaccharidosis.

Xylosides for Treatment of Mucopolysaccharidosis

Like the xylosides mentioned above, click xylosides are xylose sugars attached to an aryl or alkyl group through a triazole. This reaction is extremely selective, only taking place

between an azide and an alkyne in the presence of a metal catalyst, forming triazole. This is known as click chemistry. Click xylosides are effective with the priming of GAGs within cells. These xylosides affect tumor morphogenesis and aid in regulating various cellular processes. A method to synthesize different click carbohydrates quickly, only needing 2 steps from start to finish, and in high yields was needed. However, this was a challenge as the total synthesis of click xylosides, unprotected, and in a fast manner had never been done before. It was found that copper metal worked just as well as sodium ascorbate as a reducing agent while being much easier to remove after reaction completion. Xyloside(c) is reacted with a terminal alkyne, in the presence of copper(II) sulfate, and newly, copper metal turnings. This process virtually eliminates the need for workup, and purification procedures which are time-consuming.

Table 2: Hits for MPS Treatment



Lysosomal Storage Diseases (LSDs) are hereditary disorders characterized by a buildup of substrates within lysosomes. Mucopolysaccharidoses (MPS) are a subset of LSDs characterized by the build of glycosaminoglycans within the lysosome. Lysosomes hold enzymes that typically degrade these substrates. However, people with these diseases have dysfunctional enzymes causing the buildup of these substrates leading to severe disability and death. Prognoses for children diagnosed with MPS Type 1 are grave. It is common for children with this disease to have a life span of only around 10 years.

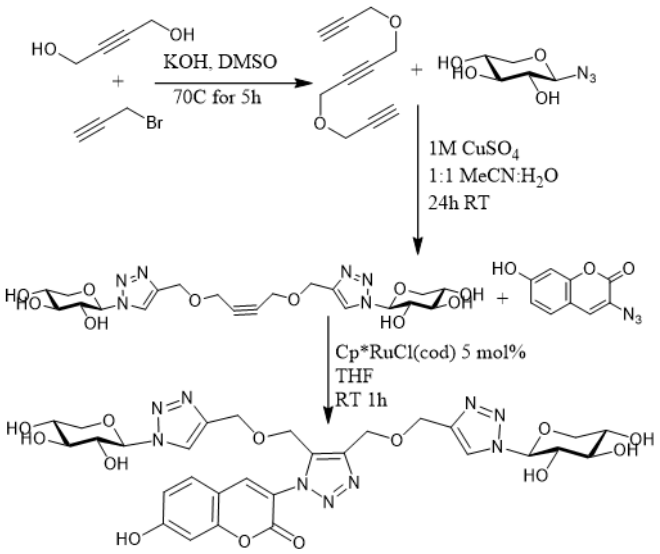
Here, xylosides(Table 2) have shown promise in addressing Mucopolysaccharidoses. In the body, GAGs are typically built off of endogenous core proteins which are uptaken and broken down by the lysosome when no longer needed. Xylosides are notable for their ability to

prime free-floating GAG chains. These GAGs, however, are excreted out of the cell, and instead of being processed by the lysosome, they are filtered out of the body through the kidney and liver. Xylosides(Table 2) were tested within cells that had this lysosome mutation and showed a significant decrease in the lysosome size after exposure. Xylosides(Table 2) were synthesized at a large scale for animal trials with our partners at CHOC and WashU, and are being further studied within vivo and human cell lines that have lysosome mutations for activity and efficacy. These 2 xylosides show little to no cytotoxicity and are orally bioavailable making them a potentially effective and noninvasive option to help mitigate this disease.

Introducing a Florescent Probe into a Bis-xylosides

Bis-xylosides prime dimerized GAG chains that interact with fibroblast growth factor receptors(FGFRs) and mediate the biological responses of fibroblast growth factors(FGFs). One key issue is that many of these small glycan-based molecules cannot be tracked inside the biological system. This can be solved by introducing a fluorophore or a probe onto the active molecule. However, previous synthetic strategies don't allow an effective way to introduce this probe into this molecule. However, with the synthetic methodologies I presented above, a combination of DMC and click chemistry can be used to synthesize these types of molecules. The idea here was to use a triyne scaffold where two xylose sugars would be clicked to the ends with CuAAC and the internal triple bond would be clicked with RuAAC.

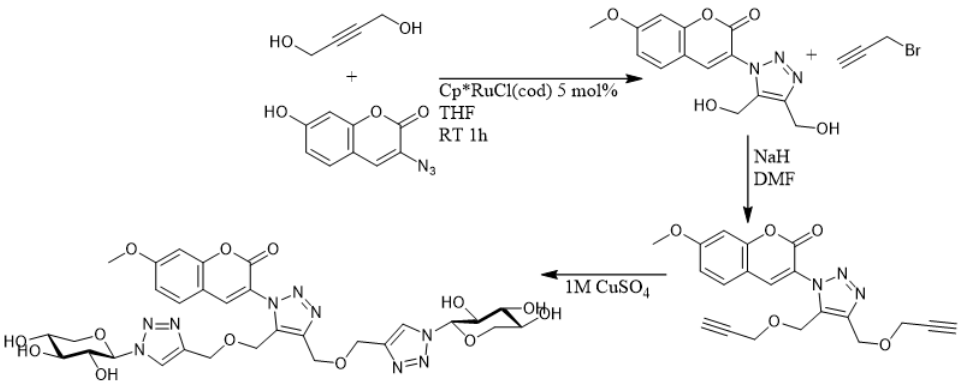
Figure 3: Route One



Initially, I tried creating a fluorophore-tagged bis-xyloside molecule by synthesizing a triyne scaffold and clicking xylose sugar to each end with CuAAC, leaving an internal triple bond intact. This internal triple bond would then

be reacted with RuAAC attaching a fluorophore and creating a disubstituted 1,4,5, triazole. However, I ran into some issues with this strategy. After introducing the two sugars at the end, the internal triple bond would not react with RuAAC. Due to this, I had to try a new strategy.

Figure 4: Route Two

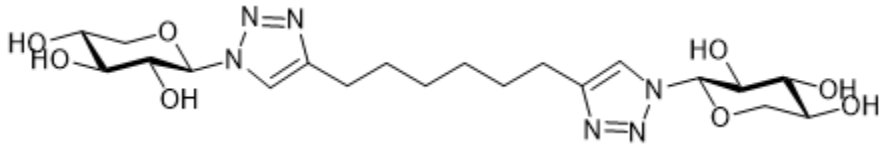
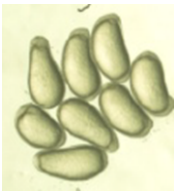


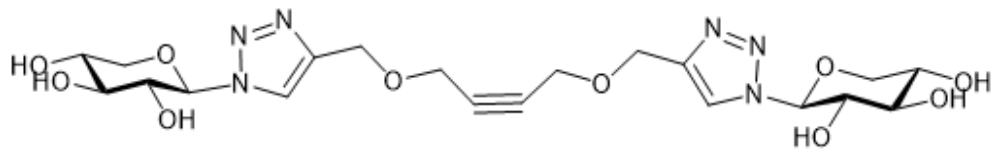
Here, RuAAC would be employed first by reacting with a diol. Then, the hydroxyls would be propargylated, and CuAAC would be used with the terminal. Here, the coumarin would be methylated instead to allow the propargylation of the hydroxyls on the diol after RuAAC.

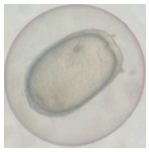
Elongation Phenotype of Bis-Xyloside

Bis-xylosides cause an elongation phenotype when injected into zebrafish embryos. This is due to FGF hyperactivation due to the resulting dimerized GAG chains that are primed after injection. To determine the possible elongation phenotype of a fluorophore-tagged xyloside, a tipple bond container bis-xyloside was injected into zebrafish embryos.

Table 2: Elongation Phenotype of Positive Control Vs Tripple Bond Bis-xyloside

	Elongation Phenotype
Positive Control 	





Discussion

The results here for the first time provide a synthetic and structural basis for the ability of a new set of bis-xylosides to promote FGF8-mediated FGFR activation. Now, these molecules can be designed with fluorophore probes or other types of probes within their structures. This synthetic base also offers a new strategy for the efficient synthesis of glycoconjugates and has been employed for the development and synthesis of new MPS therapeutics currently in animal studies. Currently, imaging of bis-xyloside fluorophore conjugates is being imaged in vivo, offering new insight into the method of action and the specific nature of interactions between GAG, FGF8, and FGFR in the formation of signaling complexes.

Experimental

Materials

Xylose, 4-nitrophenol, 4-chlorothiophenol, 3-phenyl-1-propyne, phenyl propargyl sulfide, 2-chloro-1,3-dimethylimidazolinium chloride, sodium azide, copper(II) sulfate pentahydrate, and all other materials were purchased from Sigma Aldrich or other chemical suppliers. $Cp^*RuCl(PPh_3)_2$ was purchased from Ambeed. Anhydrous organic solvents were purchased from Sigma Alrich. Deuterated solvents for NMR were purchased from Cambridge Isotope Laboratories and Oakwood Chemical. N-acetylglucoseamine was purchased from TCI. TLC was performed on MilliporeSigma™ precoated aluminum-backed silica gel 60 W F254 plates.

General Procedure A: Glycosylation of 2-Hydroxy Sugar

2-Hydroxy Sugar(1 equiv.), substrate(11 equiv.), triethylamine(28 equiv.), was added to H_2O(150mL for 6g of 2-Hydroxy Sugar) and stirred vigorously until all solutes were dissolved. The reaction mixture was then cooled to 0℃ and 2-chloro-1,3-dimethylimidazolinium chloride(3 equiv.) was added. The reaction mixture was left stirring at this temperature for 2h after which 20% ammonium hydroxide was added to the reaction mixture until neutralization of all triethylamine. The reaction mixture was then concentrated and 20% ammonium hydroxide was added again to neutralize any remaining triethylamine and then evaporated off via Rotovap. Finally, purification of the residue by silica gel flash column chromatography through a CombiFlash NextGen system gave the pure product.

General Procedure B: Glycosylation of 2-Acetamido-2-Deoxy Sugar

2-acetamido-2-deoxy sugar (1 equiv.), sodium azide(11 equiv.), triethylamine(28 equiv), was added to stirring H_2O. The reaction mixture was then lowered to 0℃ via ice bath after which 2-chloro-1,3-dimethylimidazolinium chloride(3 equiv.) was added to the reaction mixture and left stirring at this temperature for 1 hour. After this time T.L.C (typically 4:1 dichloromethane/MeOH) indicated starting material consumption and product formation with a single spot at Rf 0.4. The reaction mixture was then lowered to ~PH 2 by adding 1.2M HCl and then neutralized with a saturated sodium bicarbonate solution. The reaction mixture was then concentrated and redissolved in 20% ammonium hydroxide to neutralize any triethylamine. The solution was washed with dichloromethane 3x. The aqueous layer was then concentrated and redissolved in ethanol and filtered through a pad of Celite® 545 to remove any insolubilities. Finally, purification of the residue by silica gel flash column chromatography through a CombiFlash NextGen system gave the pure product.

General Procedure C: Glycosylation Through CuAAC

Glycosyl Azide(1 equiv.), terminal acetylene containing compound(1.1 equiv.), $CuSO_4$ * $5H_2O$ as a 1M aq solution(11 equiv.), copper metal turning(~20mg), added to a mixture of 5:3:2 t-BuOH, H_2O, EtOH. Stirred at 50 degrees Celsius in an oil bath for 12 hours after which consumption of starting material was confirmed by T.L.C dichloromethane/MeOH solvent system(typically 9:1). After this time, the reaction mixture was then concentrated via rotavap and then redissolved in dichloromethane (~20ml). The mixture was then decanted into a 25mL centrifuge tube and centrifuged to precipitate insolubilities and then decanted into a round bottom flask, removing the insolubility. The resulting dichloromethane solution was then concentrated via Rotavap to obtain a pure product.

General Procedure D: Microwave-assisted RuAAC of Internal Alkynes of Unprotected Multivalent Xylosides

Internal alkyne containing xyloside scaffold (1 equiv.), Azide containing probe(1 equiv.), $Cp^*RuCl(PPh_3)_2$ (5 mol %) was added to DMF (1 mL) and placed into a microwave vial along with a small stir bar. The reaction mixture was then subjected to microwave radiation for 4 minutes at 100℃ in a Biotage Initiator Microwave Synthesizer. After this time, T.L.C was performed to determine reaction completion $CHCl_3$/MeOH(typically 1:3). Toluene was then added to the reaction mixture and dried using a rotary evaporator. The compound was purified by Preparative HPLC through C18 Column MeOH/H_2O mobile phase in a linear gradient.

General Procedure E: RuAAC of Internal Alkynes of Unprotected Multivalent Xylosides

Internal alkyne containing xyloside scaffold (1 equiv.), Azide containing probe(1 equiv.), Cp*RuCl(COD) (5 mol %) was added to DMF (1 mL) and placed into a 5 mL round bottom flask. The reaction mixture was then left stirring at room temperature for 24 hours. After this time, T.L.C was performed to determine reaction completion $CHCl_3$/MeOH(typically 1:3). Toluene was then added to the reaction mixture and dried using a rotary evaporator. The compound was

purified by Preparative HPLC through C18 Column MeOH/H_2O mobile phase in a linear gradient.

General Procedure F: Polyyne Backbone Synthesis:

DIbromo spacer(1 equiv.) and diol internal triple bond spacer(1.1 equiv.) were added to DMSO (10 mL), and stirred for 30 minutes at room temperature, after which the reaction mixture was cooled to ℃. NaH(2.5 equiv. 60 % dispersion in mineral oil) was then added and stirred at room temperature until the desired polymer length was formed(1,4,6,12 hours) and propargyl bromide(1.2 equiv) was added to the reaction mixture to cap the polymer. The resulting Polyyne was then purified through size exclusion chromatography.

General procedure for screening xylosides:

To determine whether the xylosides can prime glycosaminoglycans in cells in vitro experiments were performed using CHO pgsA-745. The cells were treated with xylosides in the presence of 35S-Na_2SO_4, and glycosaminoglycans were purified and analyzed as described below. 1 x 10^4 cells were plated per well in Ham's/F12 complete growth medium in a 24-well plate. The cells were incubated at 37 °C in a humidified incubator for 24 h to confluency. The cells were washed with sterile PBS and replaced with 450 µL ham's/F12 containing 10% dialyzed FBS. Solutions containing a specific primer at 100x the final concentration were prepared. 5 µL of appropriate 100X primer was added to various wells to yield a final concentration of 100 µM. 50 µCi of 35S-Na_2SO_4 was also added to each well as a tracer. The 24-well plates were placed back in the 37 °C incubator for 24 h before the addition of 100 µL 6x pronase solution followed by incubation at 37 °C overnight.

Purification and quantification of primed GAGs:

The entire contents of the wells were transferred to a microcentrifuge tube and subjected to centrifugation at 16,000G for 5 min. The supernatant was transferred to a fresh tube and half a volume of 0.016% Triton X100 was added. The diluted supernatant was loaded on 0.2 mL DEAE-sepharose column pre-equilibrated with 2 mL of 20 mM NaOAc buffer pH 6.0 containing 0.1 M NaCl and 0.01% Triton X-100 and the column was washed with 4 mL of buffer described above. The bound HS/CS was eluted using 1.2 mL elution buffer, 20 mM NaOAc, pH 6.0 containing 1 M NaCl. The extent of priming by the various xyloside primers was evaluated by quantifying ^{35}S-radioactivity incorporated into the purified HS/CS elute by liquid scintillation. 50 µL of the various elutes were added to 5 mL of scintillation cocktail and the vials were counted in a scintillation counter in triplicate. The amount of radioactivity corresponds to the total glycosaminoglycan synthesized by the primer.

HPLC analysis of primed GAGs:

The purified glycosaminoglycans were analyzed by HPLC with an inline radio detector. 50 µl of elute was diluted five times with the HPLC solvent A (10 mM KH2PO4, pH 6.0, 0.2 % CHAPS) and loaded onto a weak anion exchange column, DEAE-3SW (TosoHaas Inc.,) and analyzed with the following elution profile. The sample was eluted from the column with a linear gradient of 0.2 M NaCl at 0 min to 1 M NaCl at 80 minutes at a flow rate of 1 ml per min. The radioactive glycosaminoglycans were detected by a Radiomatic flo-one A505A radio-chromatography detector. The HPLC effluent was mixed with Ultima-Flo AP scintillation cocktail in a 2:1 ratio and detected in the flow scintillation detector.

Screening of Multivalent Xylosides in Zebrafish Embryos

Zebrafish were raised and bred according to standard procedures. Strains used were Tübingen wildtypes and Tg(TOP:dGFP)w. Zebrafish experiments were approved by the University of Utah Institutional Animal Care and Use Committee. Xylosides were dissolved in

distilled water to a final concentration of 100 mM for injection into embryos. Embryos were raised at 28.5 °C in an E3 buffer. Xylosides were diluted with 1% phenol red, and each Xyloside (33 pmoles) was microinjected in a 1 nL bolus into the blastomere of one-cell stage embryos. Embryos were staged, fluorescent imaging was taken periodically of embryos, and their phenotypes were compared to those of wild-type embryos.

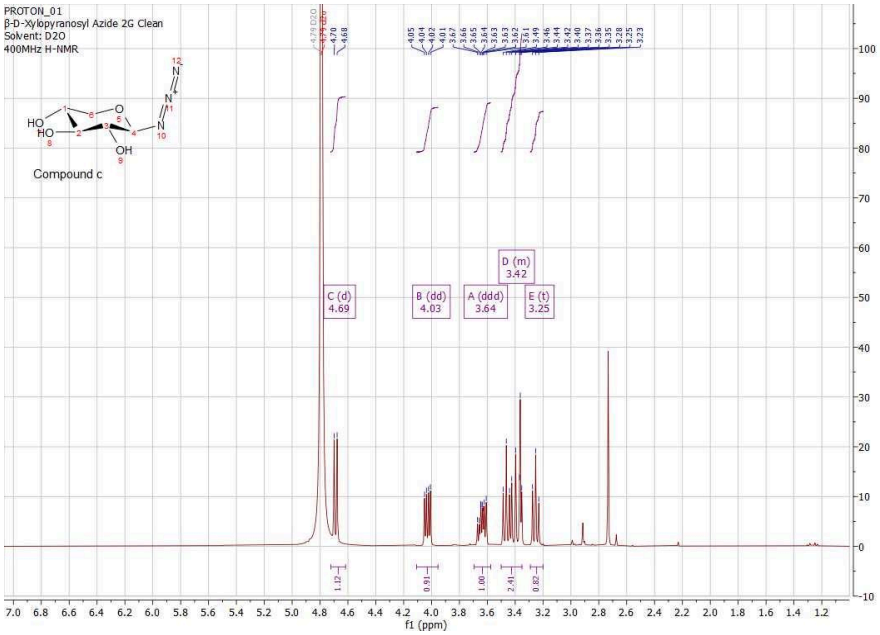
Recapturing Multivalent Xyloside Biotin Conjugates

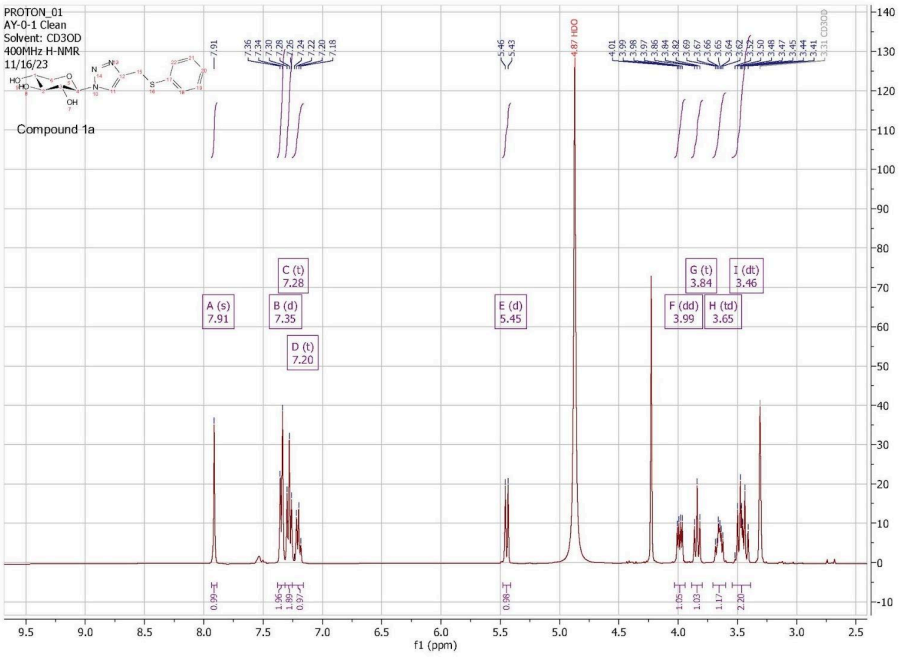
At the one-cell stage, 100 embryos were injected with biotinylated xyloside, then lyophilized at 12 hours post-fertilization (hpf), and treated overnight with 0.016% pronase at 37°C. After centrifugation, the supernatant was mixed with an equal volume of 0.016% Triton X-100 and applied to a DEAE-sepharose column (0.2 ml) pre-equilibrated with wash buffer (20 mM NaOAc, 0.1 M NaCl, and 0.01% Triton X-100, pH 6.0). The column was washed with 6 ml of wash buffer, and the bound glycosaminoglycans (GAGs) were eluted using elution buffer (20 mM NaOAc, 1 M NaCl, pH 6.0; 1.2 ml). The eluate was then desalted and concentrated using 3000 MWCO Amicon columns. S-AF350 (0.2 µg) was added, and the mixture was left to incubate overnight at room temperature in the dark to capture the multivalent xyloside-primed GAGs.

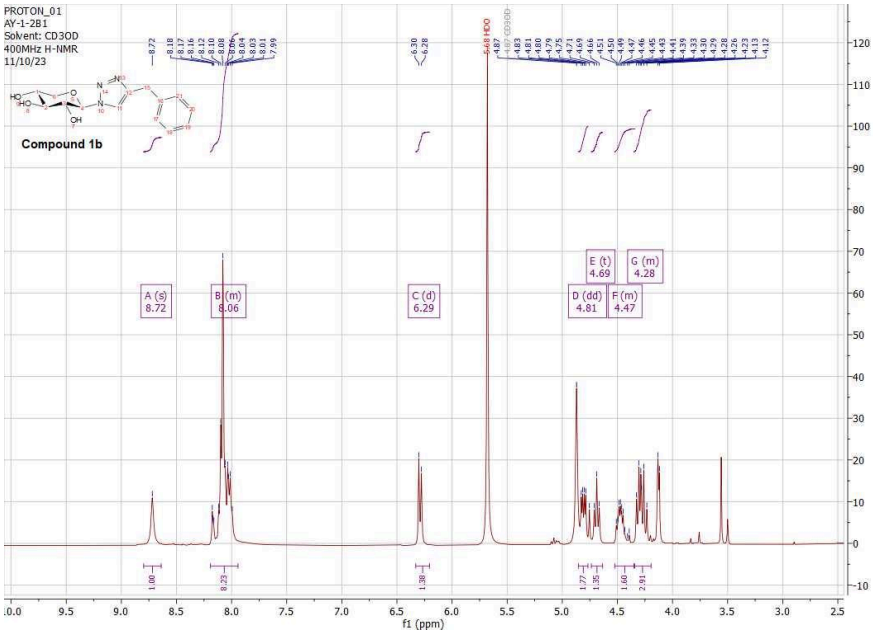
Structural and Compositional Analysis of Recaptured Multivalent Xyloside-primed GAGs.

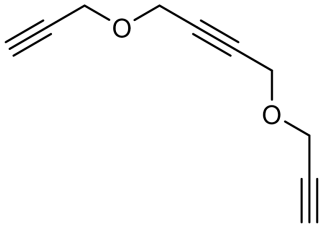
The GAG-streptavidin complexes were analyzed using a high-performance liquid chromatography (HPLC) system coupled to a fluorescent detector. The excitation and emission wavelengths were set at 346 nm and 442 nm, respectively. The GAG-streptavidin conjugate was injected onto two tandem G3000SWxl columns and eluted with a solution containing 100 mM KH2PO4, 150 mM NaCl, pH 6.0, at a flow rate of 0.5 ml/min for 60 minutes. As controls, GAGs from untreated embryos were also treated with S-AF350. Subsequently, the GAG-streptavidin conjugate was digested overnight at 37°C with heparinase I, II, and III, or chondroitinase ABC.

The digested samples were then loaded onto the same column to determine the composition of heparan sulfate (HS) and chondroitin sulfate (CS).

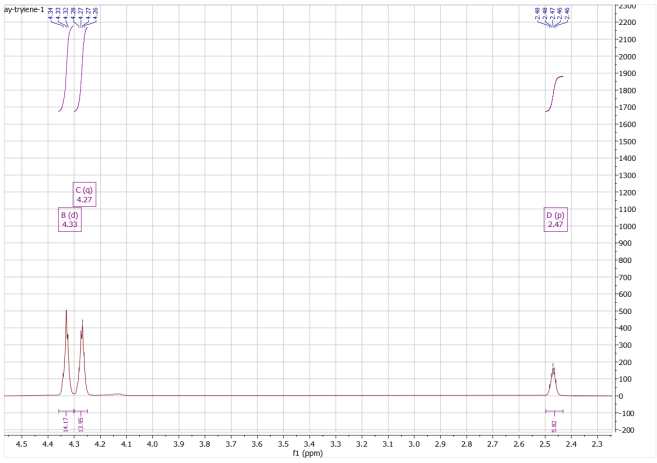




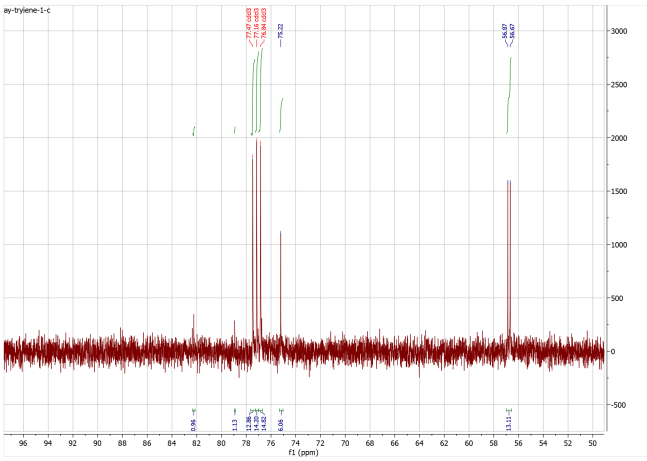




1H NMR (CDCl3, 400 MHz): δ = 2.47 (t, J = 2.4 Hz, 2H), 4.27 (d, J = 2.4 Hz, 4H), 4.33 (s, 4H).



^{13}C NMR (101 MHz, cdcl3) δ 82.3, 79, 77.47, 77.16, 76.84, 75.22, 56.87, 56.6



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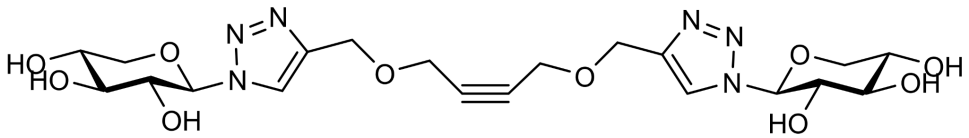
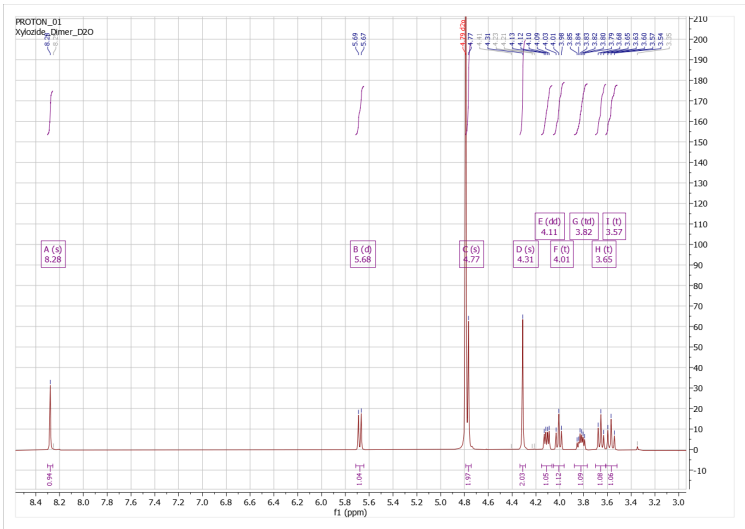
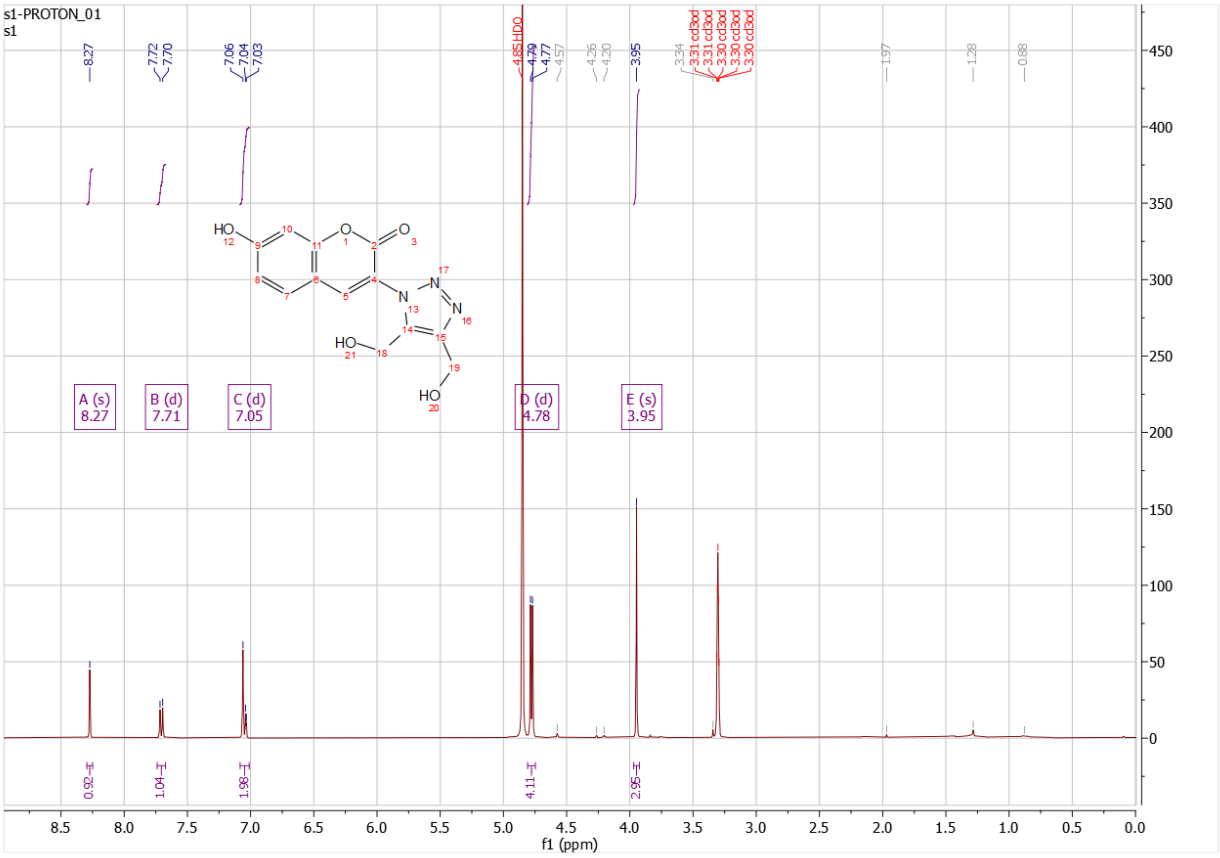


Prepared with modification of general procedure C (2 equiv xylose azide). 1H NMR (400 MHz, d2o) δ 8.28 (s, 1H), 5.68 (d, J = 9.2 Hz, 1H), 4.77 (s, 2H), 4.31 (s, 2H), 4.11 (dd, J = 11.5, 5.5 Hz, 1H), 4.01 (t, J = 9.2 Hz, 1H), 3.82 (td, J = 10.0, 5.4 Hz, 1H), 3.65 (t, J = 9.2 Hz, 1H), 3.57 (t, J = 11.0 Hz, 1H).




s1-PROTON_01
s1
A (s)
8.27
B (d)
7.71
C (d)
7.05
D (d)
4.78
E (s)
3.95
f1 (ppm)